

Hotel Booking Cancellation Analysis

Prepared By: Zain Ul Abideen

Executive Summary:

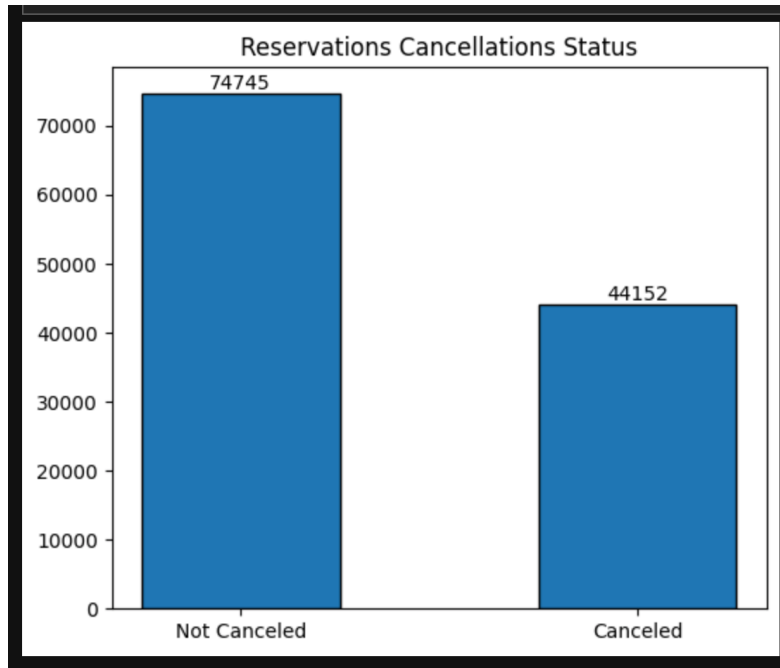
This project analyzes hotel booking data to identify the key factors influencing reservation cancellations. Using Python and SQL, the analysis explores booking trends, hotel performance, pricing behavior, customer segments, and booking channels. The findings reveal that cancellation rates vary significantly across hotel types, market segments, and customer groups, providing valuable insights for improving occupancy and revenue management.

Business Problem:

Hotel booking cancellations can negatively impact occupancy, revenue forecasting, and operational planning. The objective of this analysis is to identify cancellation patterns and uncover the factors most strongly associated with booking cancellations.

Reservation Status Analysis:

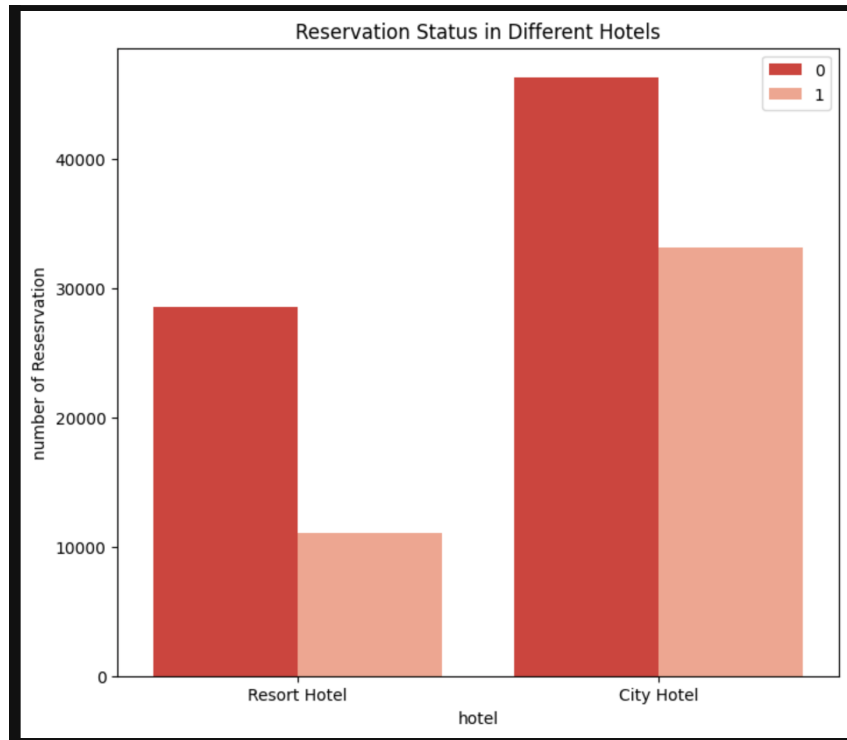
The dataset contains 74,745 successful reservations and 44,152 cancelled reservations. While most bookings were completed successfully, the volume of cancellations represents a significant operational and revenue challenge for hotels.



Hotel Type Analysis:

Cancellation behavior differs considerably between hotel types. Resort Hotels recorded a cancellation rate of approximately 27%, while City Hotels experienced a much higher cancellation rate of around 41%.

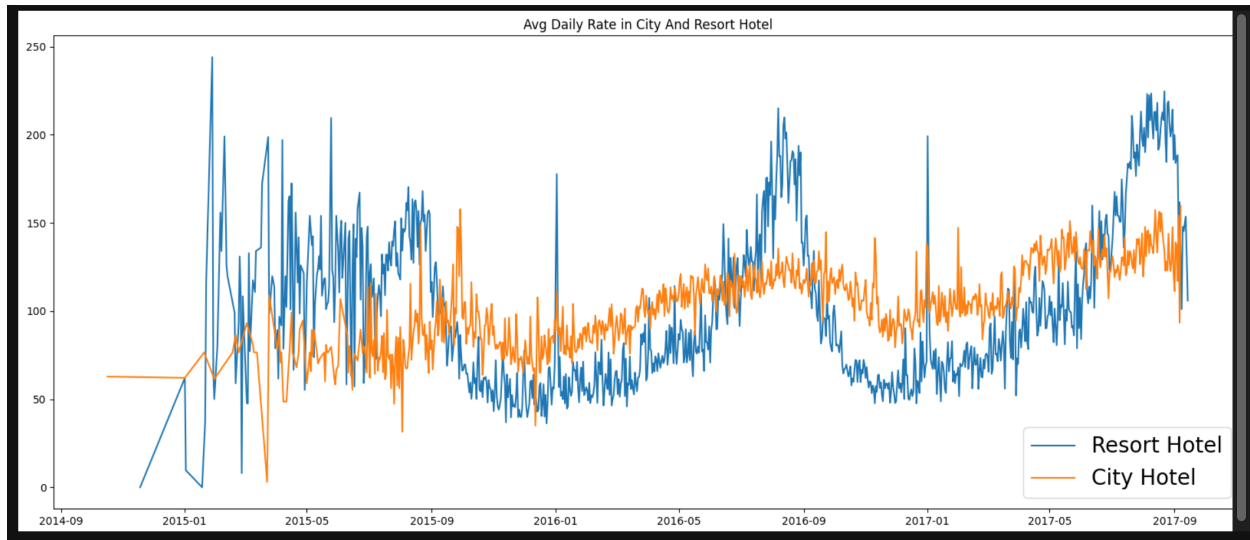
This suggests that City Hotels are more vulnerable to booking cancellations and may require stronger cancellation management strategies.



ADR Analysis by Hotel Type:

The Average Daily Rate (ADR) analysis showed that Resort Hotels generally maintain higher room rates than City Hotels. Although City Hotels occasionally recorded higher ADR values on certain dates, the highest pricing peaks were consistently observed in Resort Hotels.

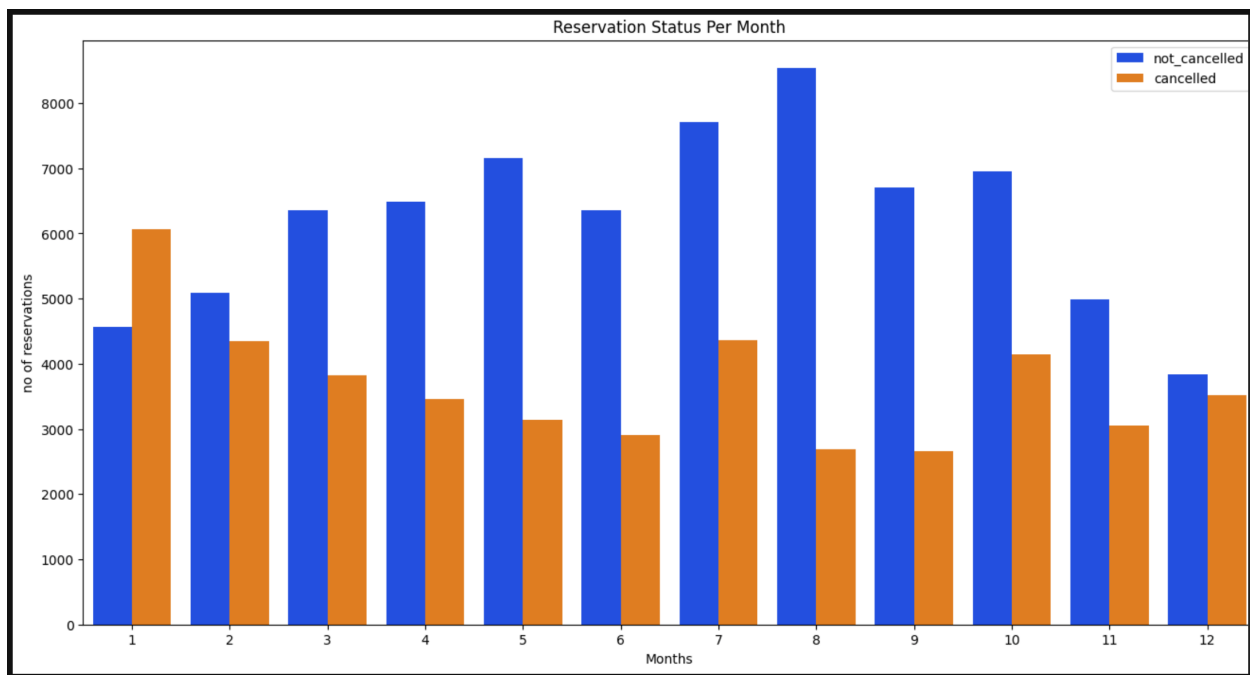
This indicates stronger pricing power and revenue potential within the Resort Hotel segment.



Monthly Reservation Trends:

Booking activity showed clear seasonal patterns throughout the year. August recorded the highest number of successful reservations, indicating strong customer demand. In contrast, January recorded the highest number of cancellations, while December showed the lowest number of successful bookings.

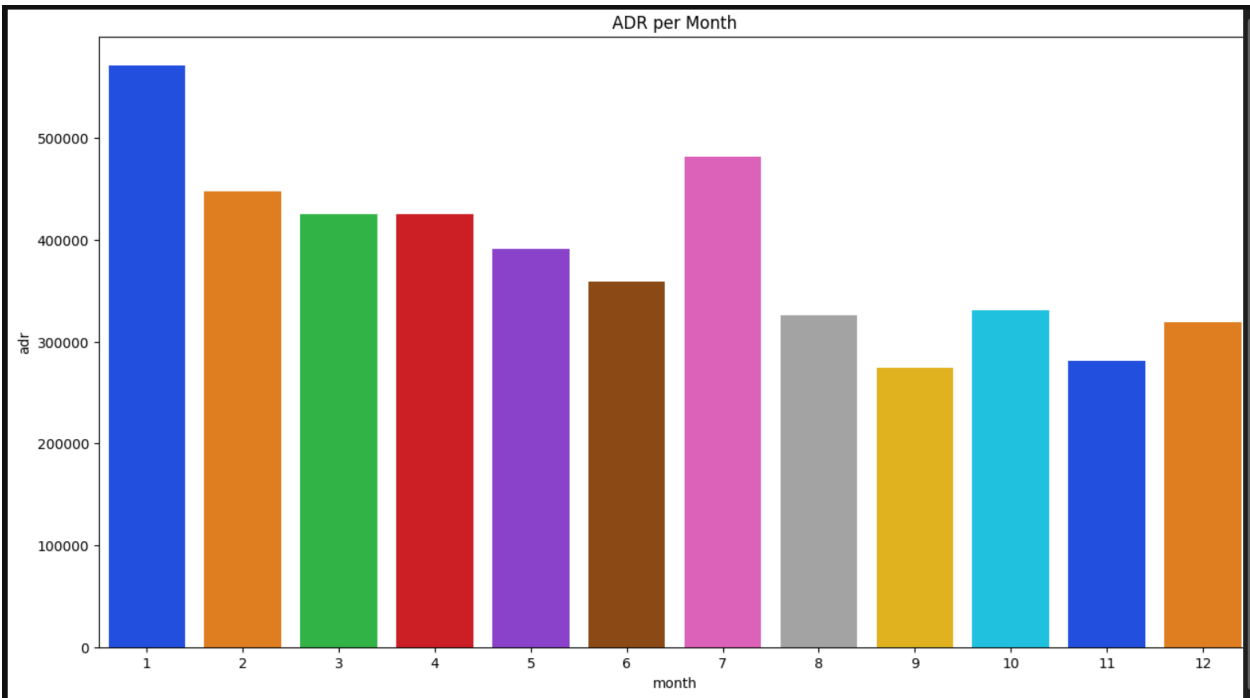
These trends highlight the importance of seasonal planning and demand forecasting.



Monthly ADR Trends:

ADR fluctuated throughout the year, with the highest values observed in January and July. September recorded the lowest ADR, indicating weaker pricing performance during that period.

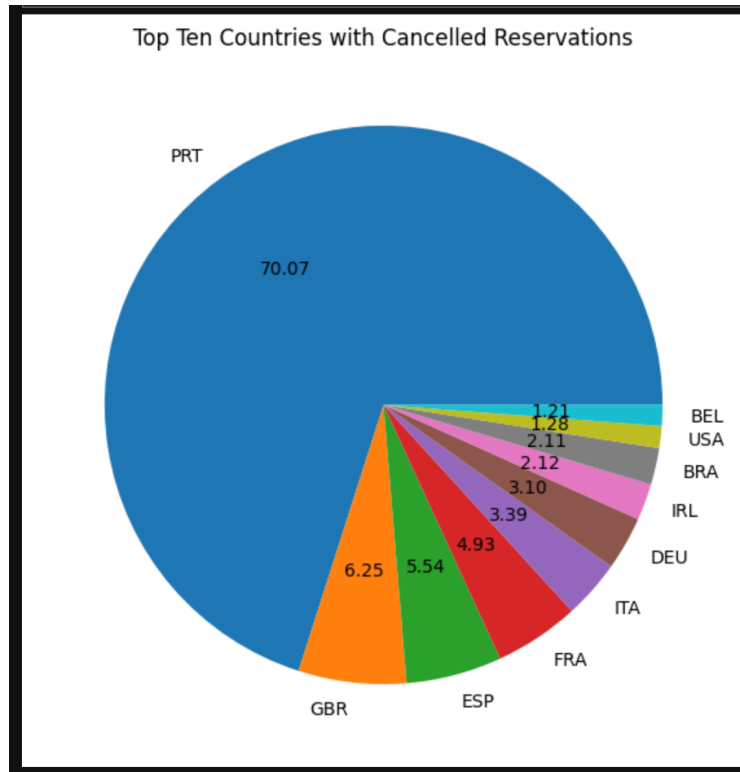
Understanding these seasonal pricing patterns can support more effective revenue management strategies.



Country-Wise Cancellation Analysis:

The analysis revealed that approximately 70% of all cancelled reservations originated from PRT (Portugal). The remaining cancellations were distributed across countries such as GBR, ESP, FRA, ITA, DEU, BRA, USA, and BEL.

This unusually high concentration suggests that customer behavior within the Portuguese market may have a significant impact on overall cancellation performance.



Market Segment Analysis:

Online Travel Agencies (Online TA) generated the largest share of reservations, accounting for approximately 47% of all bookings. Offline Travel Agencies, Groups, Direct bookings, and Corporate bookings contributed smaller portions.

The results indicate a strong dependence on online booking platforms for customer acquisition.

Market Segment Cancellation Analysis:

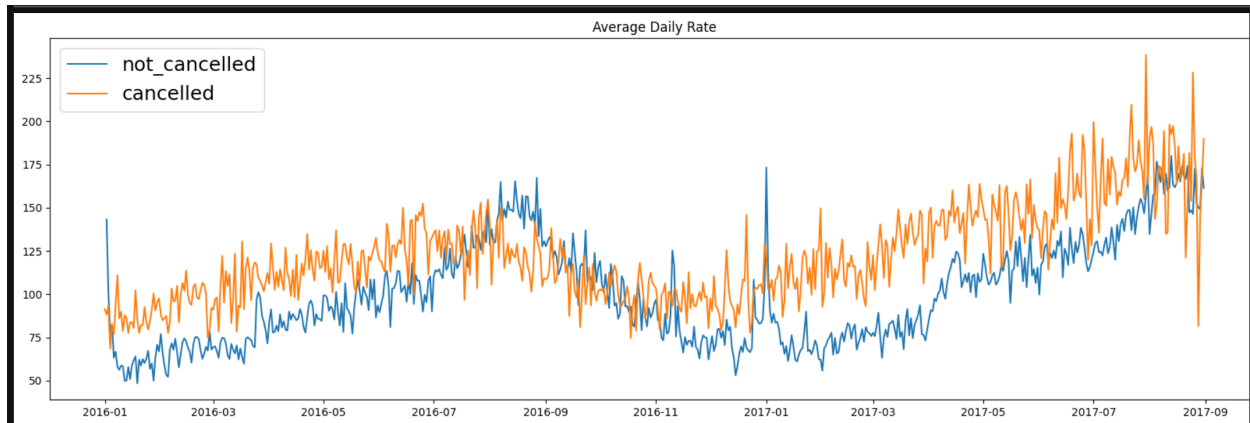
Online Travel Agencies also generated the highest share of cancellations, contributing approximately 46% of all cancelled bookings. Group reservations accounted for 27% of cancellations, followed by Offline Travel Agencies at 18%.

These segments represent the largest sources of cancellation risk within the dataset.

ADR and Cancellation Relationship:

A comparison between cancelled and non-cancelled reservations revealed that cancelled bookings generally had higher ADR values. This suggests that higher-priced reservations are more likely to be cancelled than lower-priced bookings.

Since these bookings generate higher revenue, their cancellation can have a greater financial impact on hotel performance.



Key Insights:

- City Hotels experience significantly higher cancellation rates than Resort Hotels.
- Resort Hotels maintain higher ADR values and stronger pricing performance.
- August recorded the highest successful booking volume.
- January recorded the highest cancellation volume.
- Portugal accounted for approximately 70% of all cancellations.
- Online Travel Agencies generated both the highest booking volume and the highest cancellation volume.
- Higher ADR bookings showed a greater tendency to be cancelled.

Recommendations:

- Prioritize cancellation reduction strategies for City Hotels.
- Investigate the unusually high cancellation rate from Portugal.
- Strengthen direct booking channels to reduce dependency on third-party platforms.
- Review cancellation policies associated with Online Travel Agencies and Group bookings.
- Monitor high-ADR reservations more closely, as they carry a higher cancellation risk.
- Use seasonal booking and pricing trends to optimize revenue management and occupancy planning.

Conclusion:

The analysis demonstrates that hotel booking cancellations are influenced by hotel type, booking channel, customer location, seasonality, and pricing behavior. City Hotels, Online Travel Agencies, and high-ADR reservations emerged as the most significant risk areas. By addressing these factors through targeted pricing, channel management, and cancellation reduction strategies, hotels can improve occupancy rates, reduce revenue loss, and enhance overall business performance.